

Programming Life for Practical Solutions

Synthetic Biology for a Sustainable Future

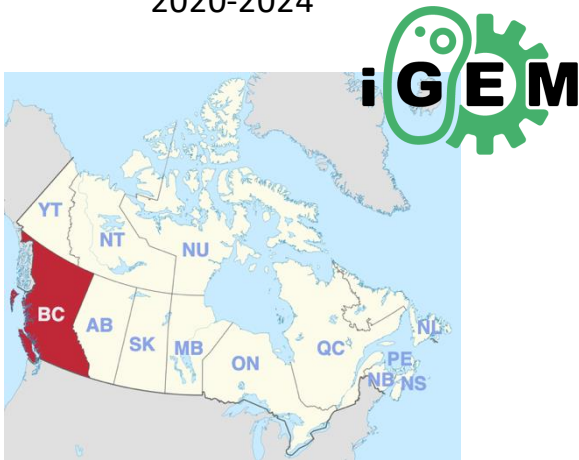
Created and presented for the CTU by Trevor David Fox
September 27, 2026



About Me



University of British Columbia, Canada
B.Sc. Biochemistry and Med. Mol. Biol.
Thesis in Analytical Biochemistry
2020-2024



UNIVERSITÉ ÉVRY
PARIS-SACLAY

University of Évre Paris-Saclay, France
M.Sc. Synthetic Biology

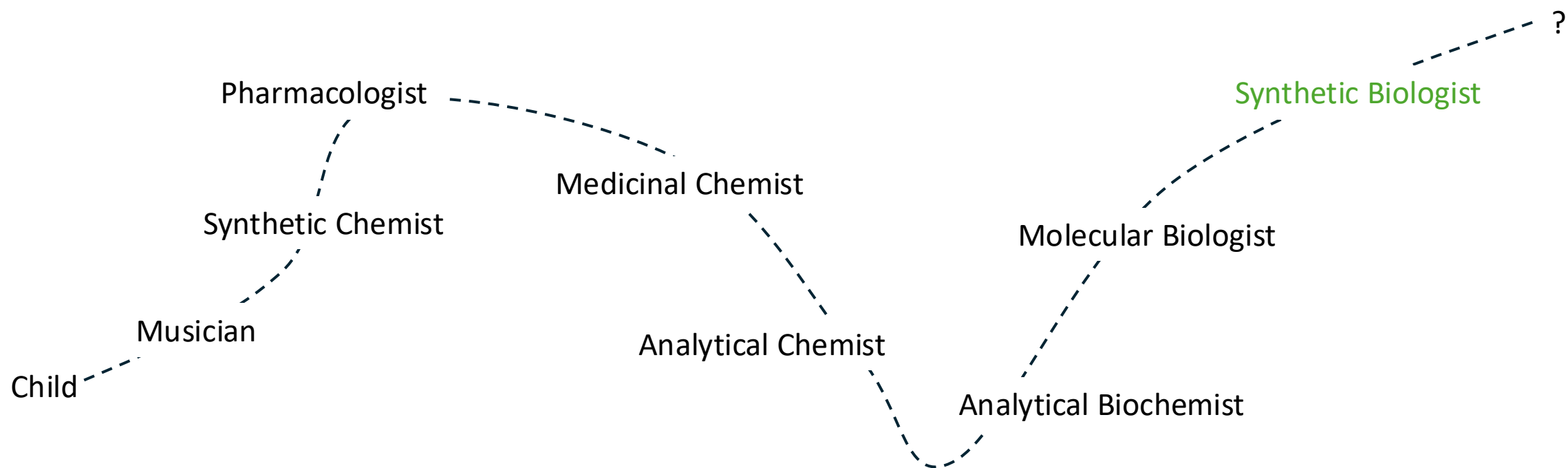
2025-Current



?

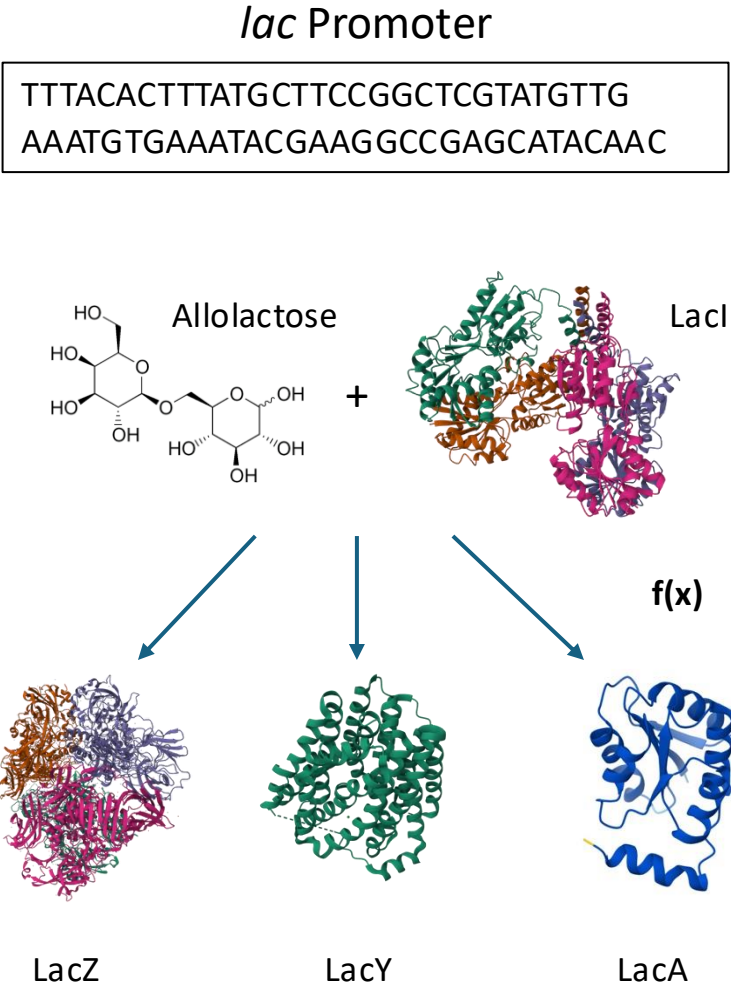
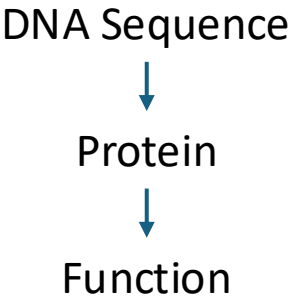
The life journey...

On the journey...



First... Synthetic Biology??

```
def lacOperon(allolactose = False, LacI = False):  
    if allolactose and LacI:  
        produceProtein("LacZ")  
        produceProtein("LacY")  
        produceProtein("LacA")
```



Where does it come from?

Genetic Engineering + Mol. Bio.

- Techniques

Systems Biology

- Decomposing Complex Systems

Engineering

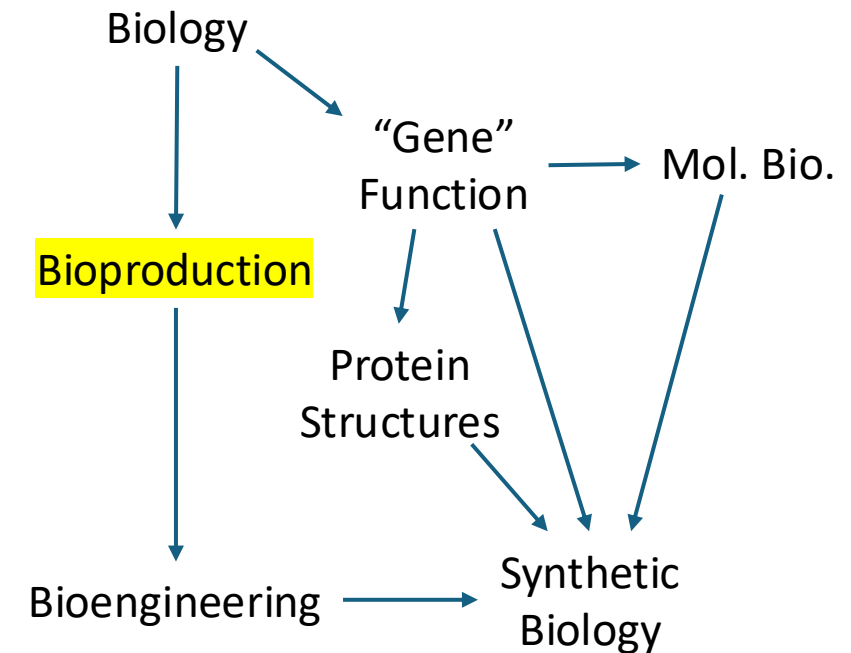
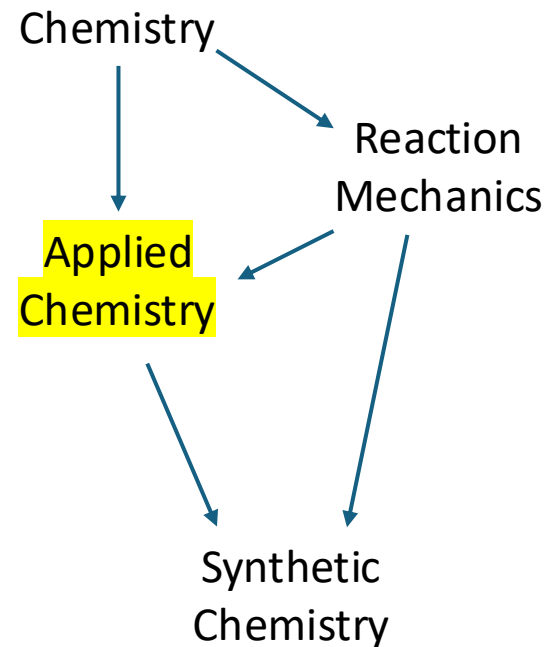
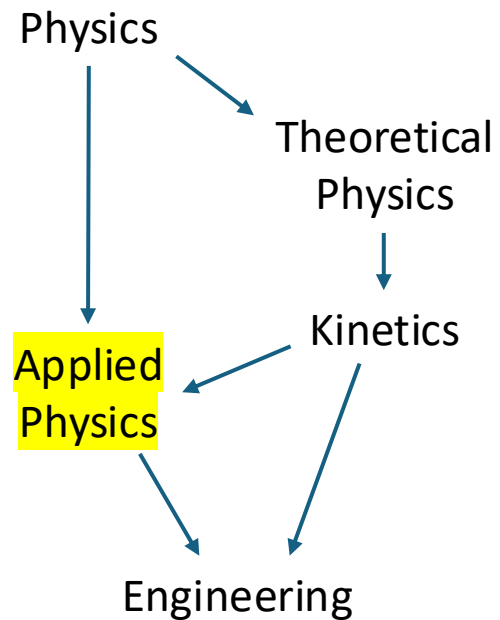
- Principles and Language

Synthetic Biology

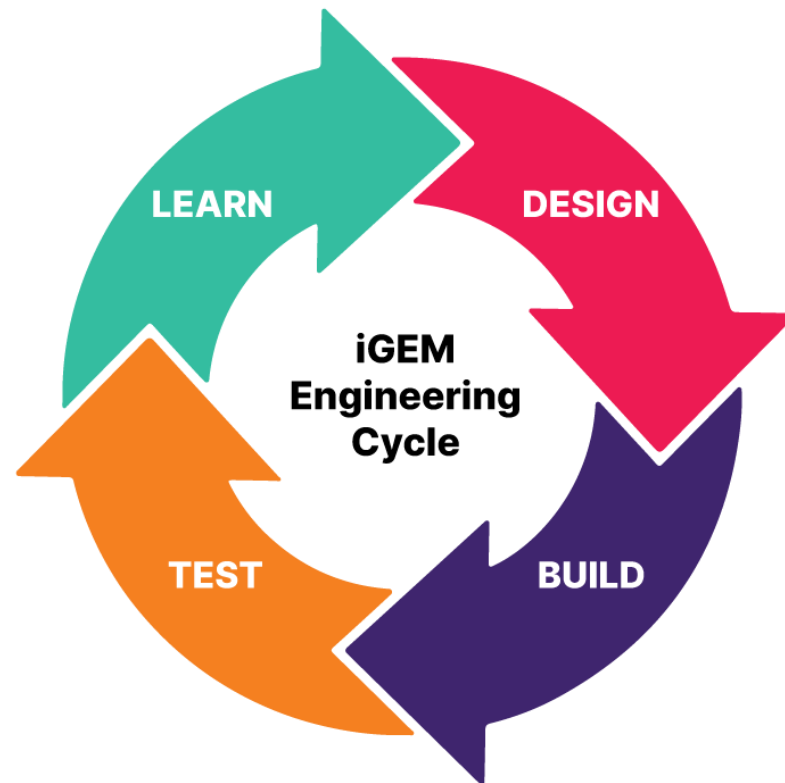
- Bottom-Up Approach
- Forward and Reverse Engineering

“What I cannot create, I do not understand”
– R. Feynman

Where does it come from?



Engineering in SynBio



Every time you turn the loop, your design becomes better.

We learn about the system **we create**.
Not about what **nature made before us**.

Vocabulary: Chassis, Part, Device, Circuit, Assembly,
Vehicle, Sensor, Actuator, ...

Standard Parts



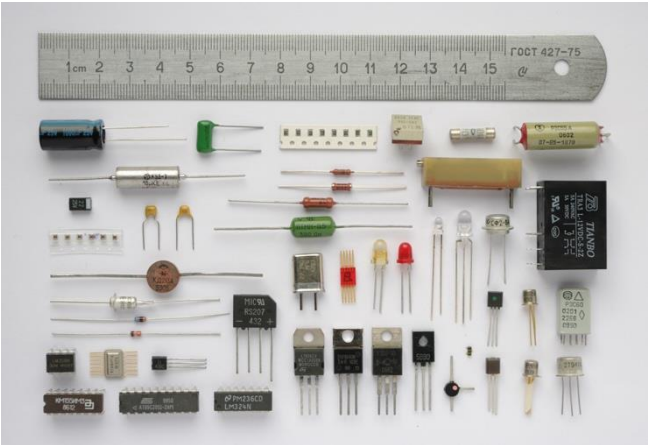
Henry Maudslay, 1797



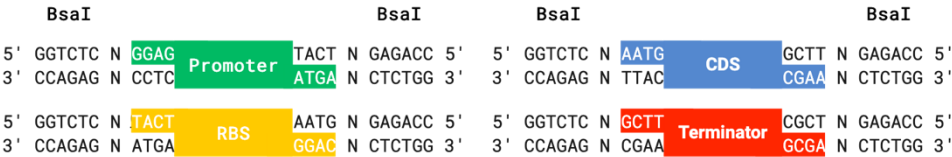
Hans Camenzind, 1971



David Baker, 2023



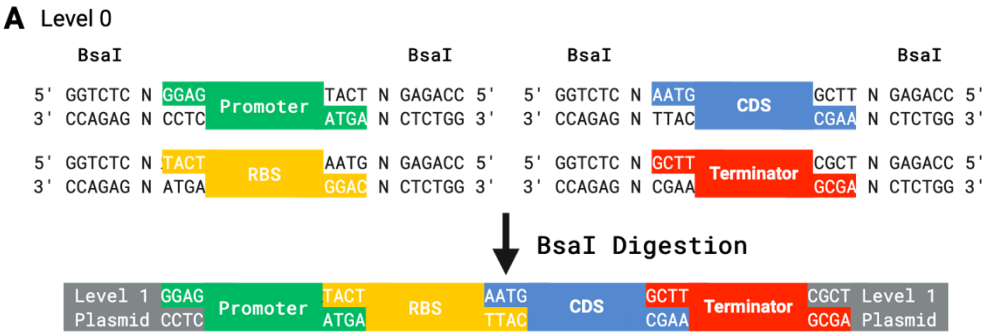
A Level 0



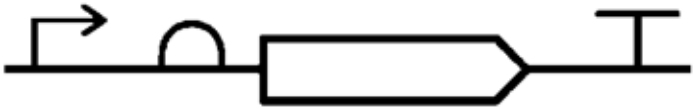
	promoter		primer binding site
	cds		restriction site
	ribosome entry site		blunt restriction site
	terminator		5'sticky restriction site
	operator		3'sticky restriction site
	insulator		5'overhang
	ribonuclease site		3'overhang
	rna stability element		assembly scar
	protease site		signature
	protein stability element		engineered region
	origin of replication		

Standard Parts

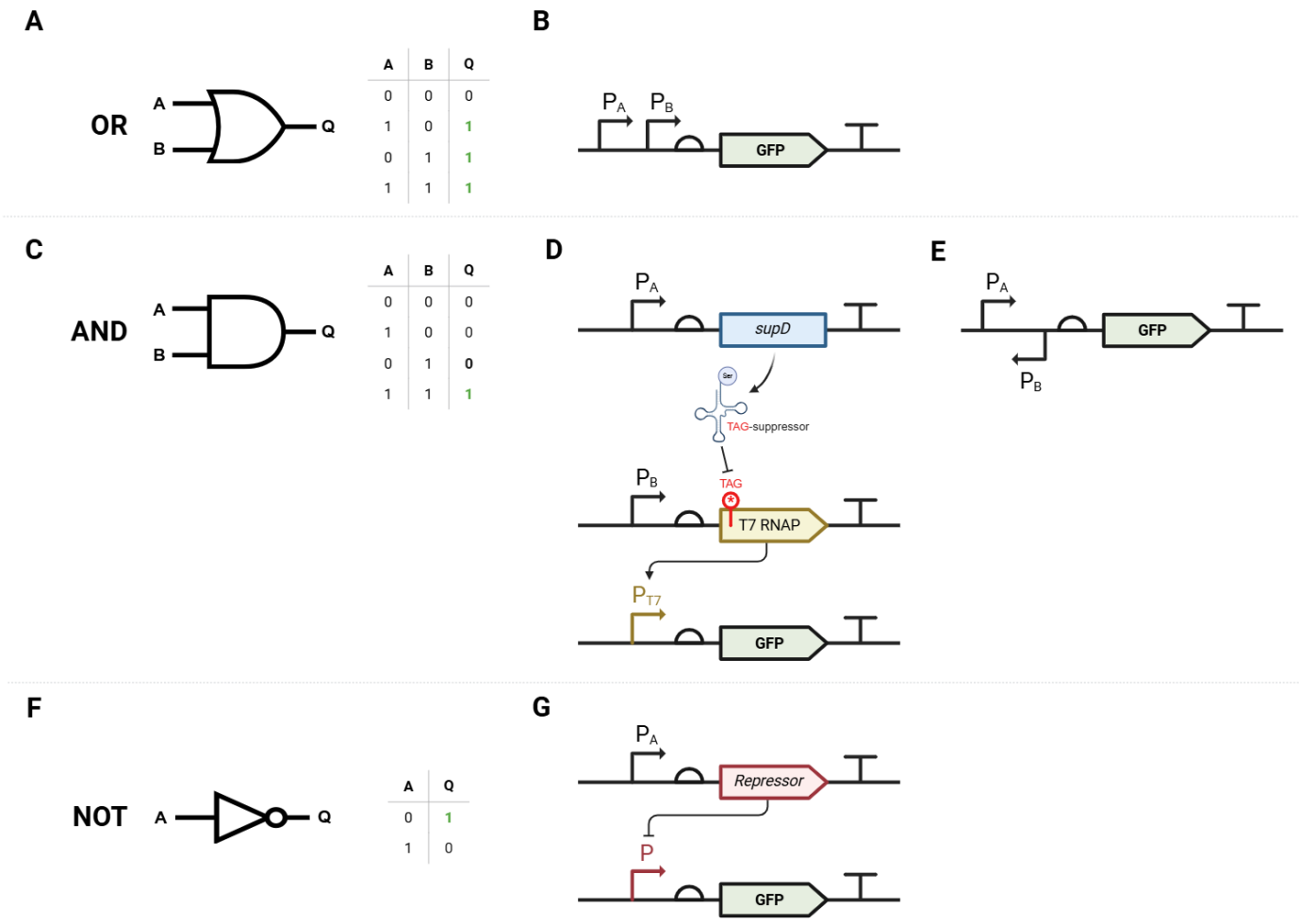
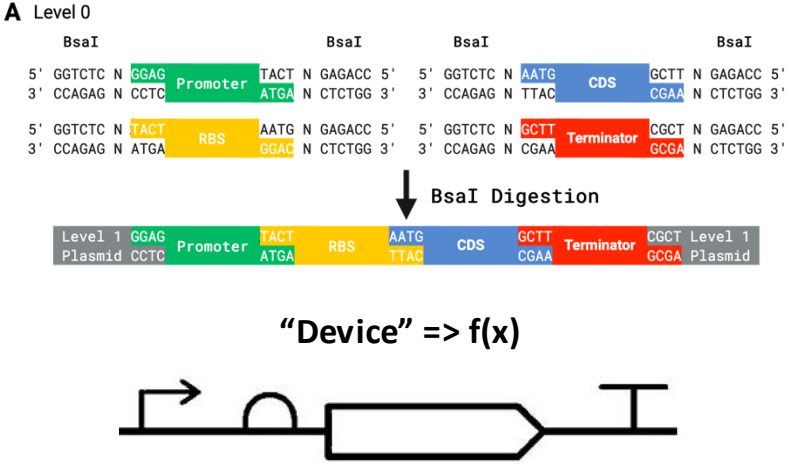
 promoter	 primer binding site
 cds	 restriction site
 ribosome entry site	 blunt restriction site
 terminator	 5'sticky restriction site
 operator	 3'sticky restriction site
 insulator	 5'overhang
 ribonuclease site	 3'overhang
 rna stability element	 assembly scar
 protease site	 signature
 protein stability element	 engineered region
 origin of replication	



“Device” => f(x)



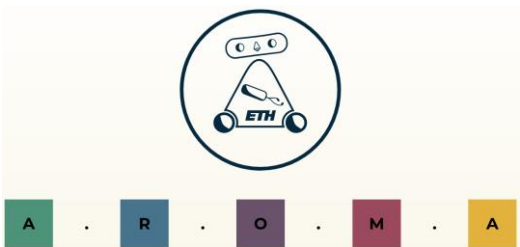
Standard Parts



Devices, Circuits, and More!



ETH Zurich



https://2018.igem.org/Team:ETH_Zurich

Chemotropic Robot!

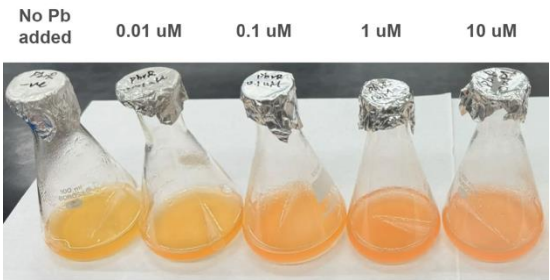


Hong Kong JSS



<https://2024.igem.wiki/hongkong-jss/>

Lead Biosensor! (and Cd, Cu, Zn)

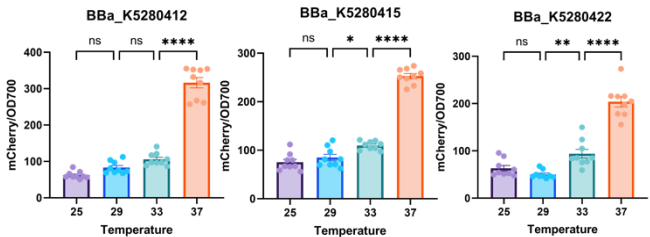


HKUST(GZ)

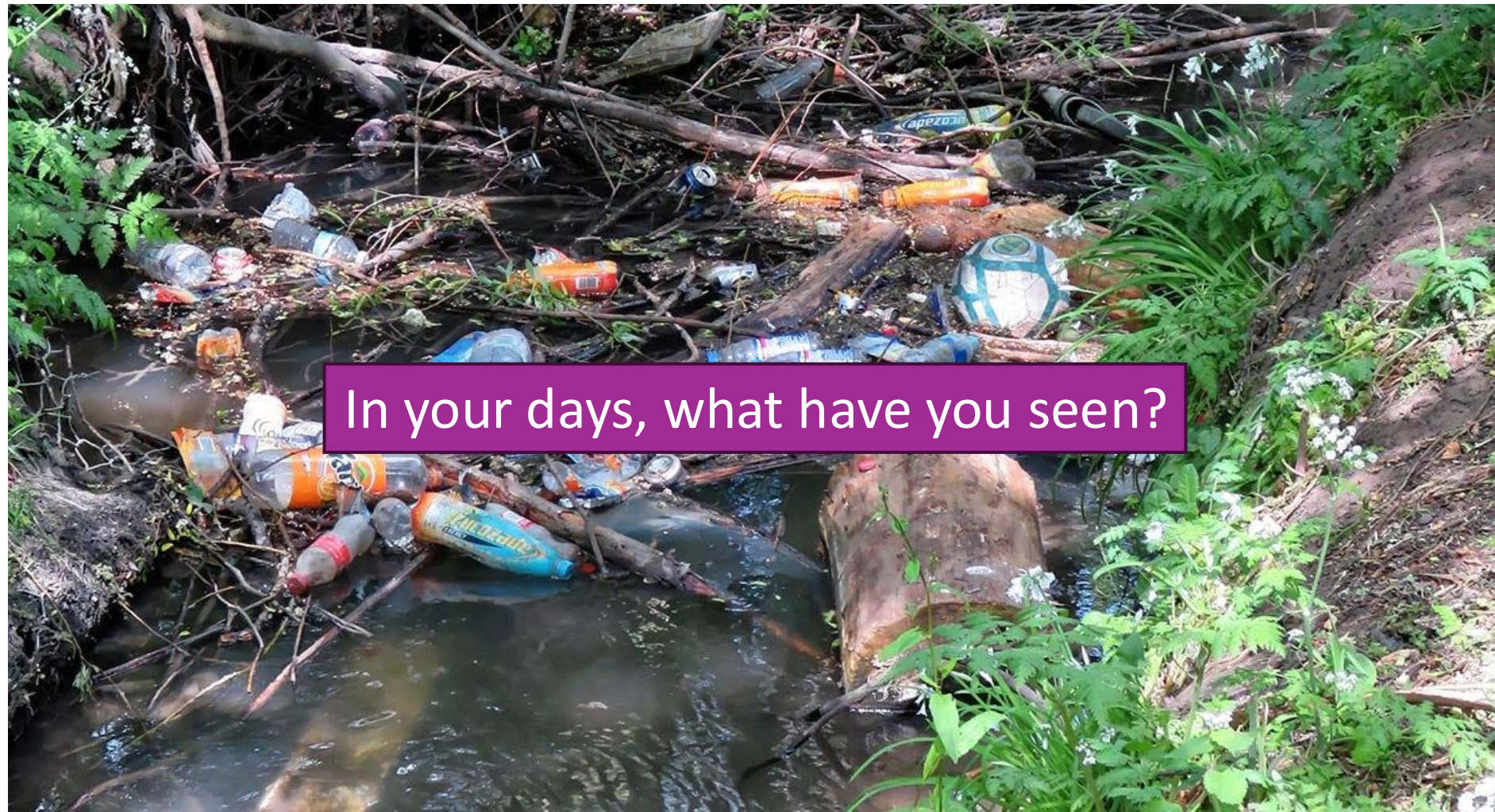


<https://2024.igem.wiki/hkust-gz/>

Thermosensor! (and Light Sensor)



Disruption of wildlife habitat



In your days, what have you seen?

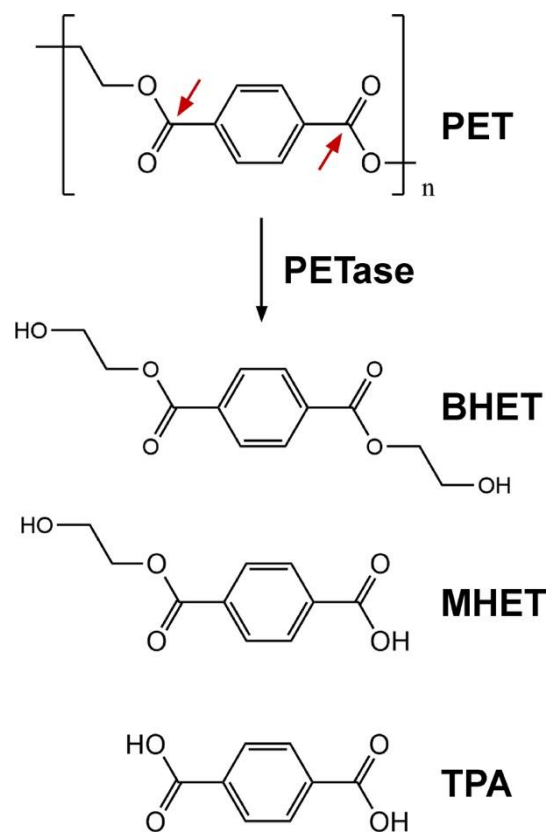
New
microbiological
niches

Microplastics
in waterways

Leaching plasticizers

What can you do?

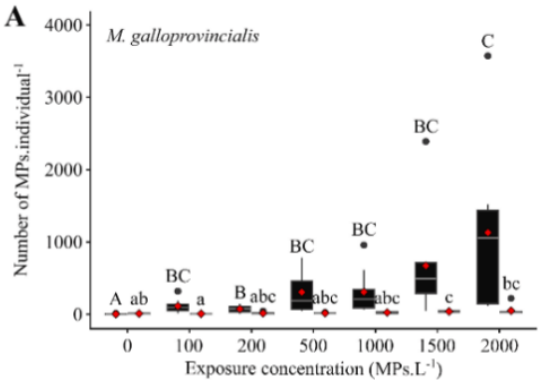
Fix It



<https://doi.org/10.1111/febs.14612>

Monitor It

<https://doi.org/10.1016/j.aquatox.2025.107659>



Use It



<https://silviapvadi.wordpress.com/2015/04/10/cyanobacteria/>

And more...

“Fix It” – Microplastic Degradation

Plastics are durable by design.

~460 million tonnes of
plastic produced per year*

(Tournier et al., 2020)



1–1.7 million tonnes of plastic
enter the ocean per year.

(OECD Global Plastic Outlook, 2022)



Over 170 trillion plastic particles
now afloat in the ocean

(Eriksen et al., 2023)

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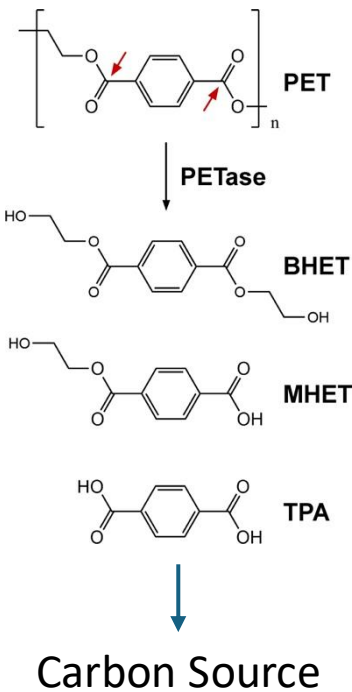
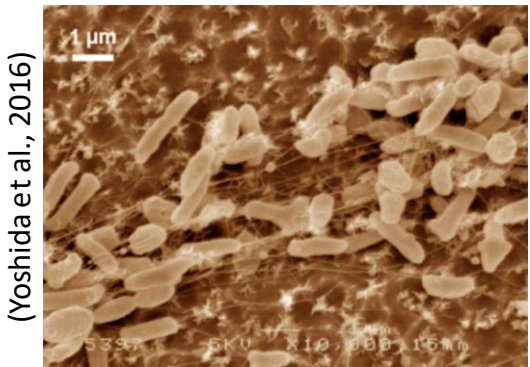
Over 170 trillion plastic particles now afloat in the ocean

(Eriksen et al., PLOS ONE, 2023)

PET degradation site in Tsuyama, Japan



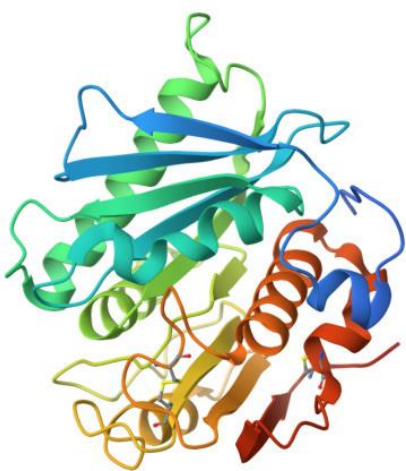
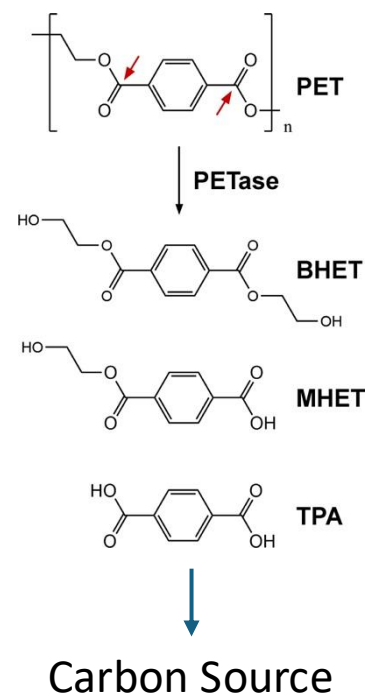
Ideonella sakaiensis



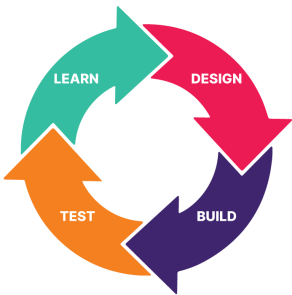
*70 million tonnes comprised by PET

“Fix It” – Microplastic Degradation

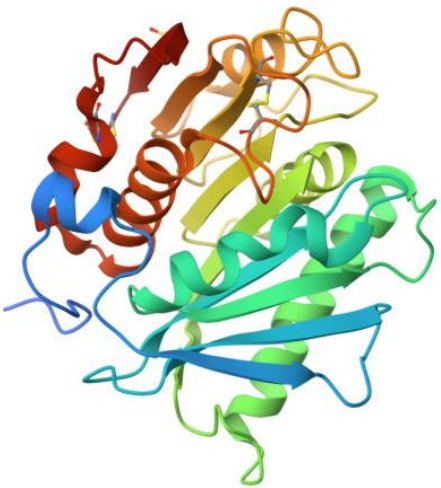
Plastics are durable by design.



I. sakaiensis PETase



BBa K5023003



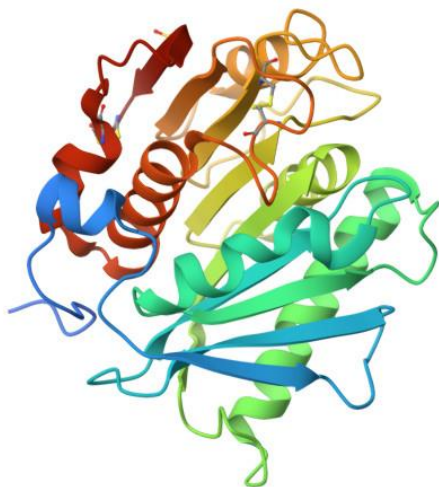
FAST-PETase

Produced by ML models:
Faster, higher thermocompatibility
(Lu et al., Nature, 2022)

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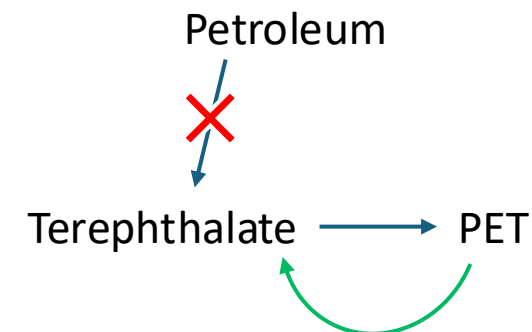
UNILA-LatAm (2023)

<https://2023.igem.wiki/unila-latam/>



Engineering microalgae (*C. reinhardtii*) with FAST-PETase
for deployment in wastewater facilities.

Imagine...



“Watch It” – Ecosystem Monitoring



Moalboal, Cebu

How do you monitor the biodiversity of a reef?

Not just the animals... the bacteria? Microalgae?

Huang et al. (2023)* barcoded
Cebu fish, identifying 393 teleost
species across 229 genera.

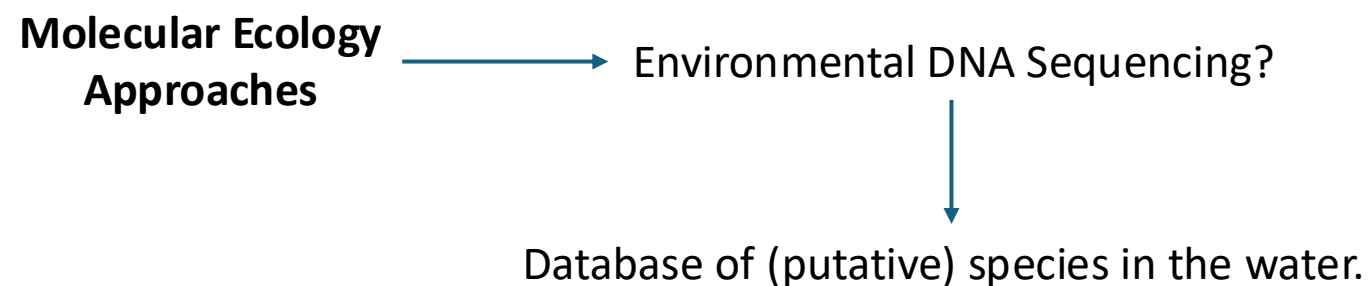


**Molecular Ecology
Approaches**

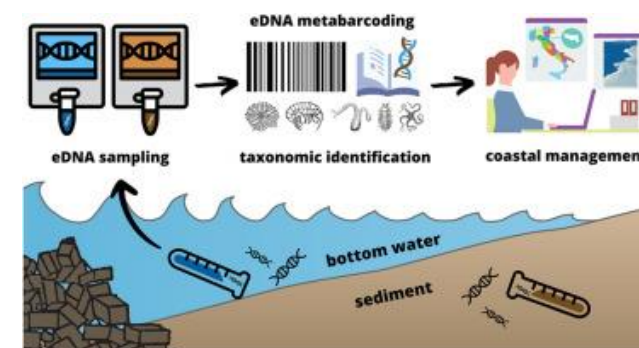
“Watch It” – Ecosystem Monitoring



Moalboal, Cebu

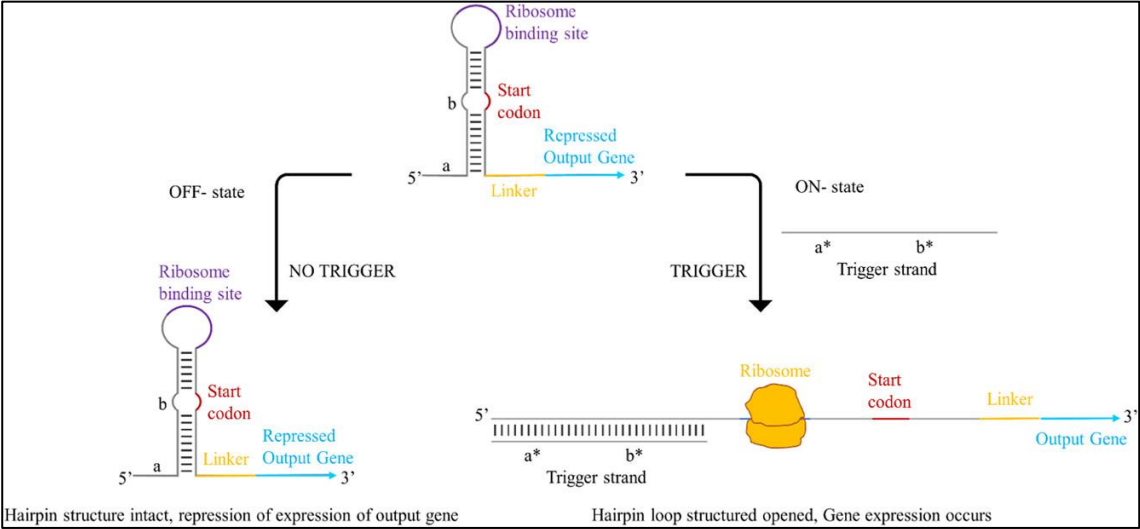


Bautista et al. (2023)* performed eDNA metabarcoding in Mindanao!
Very possible.



“Watch It” – Ecosystem Monitoring

<https://doi.org/10.1002/bit.28309>



RNA Toehold Switch

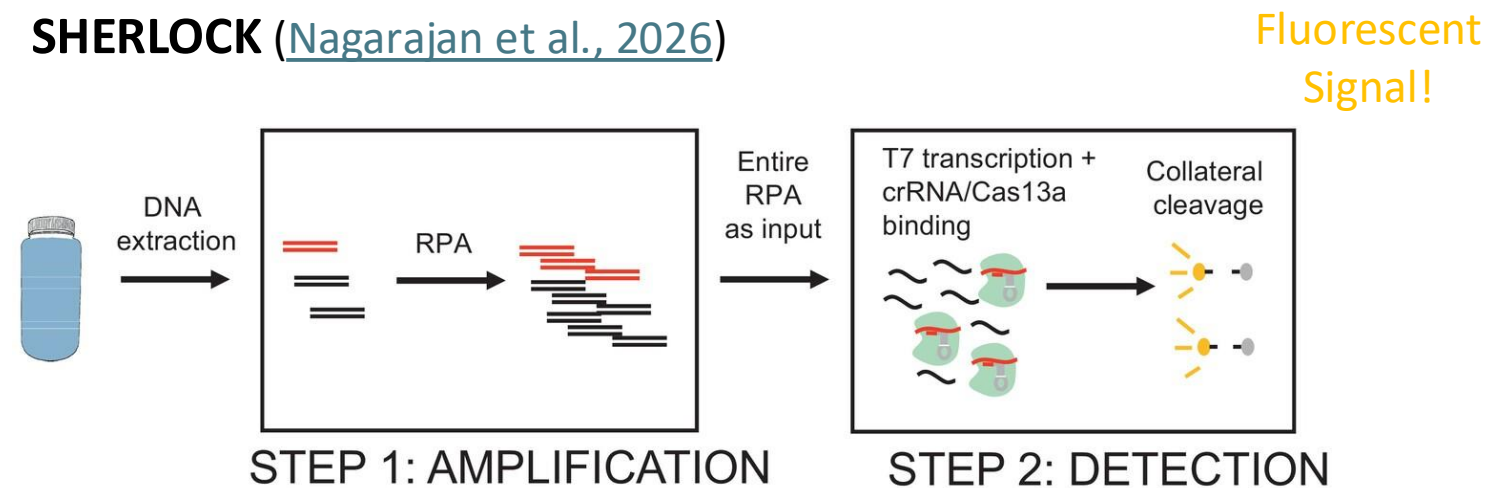
Engineered Biosensors!



“Watch It” – Ecosystem Monitoring

Even more cutting edge...

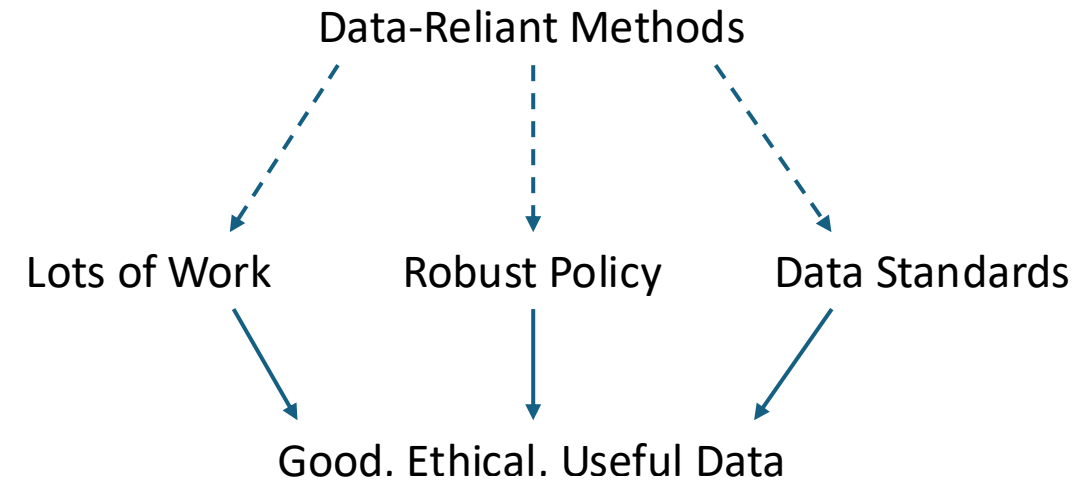
SHERLOCK ([Nagarajan et al., 2026](#))



“Watch It” – Ecosystem Monitoring

There are still some limitations!

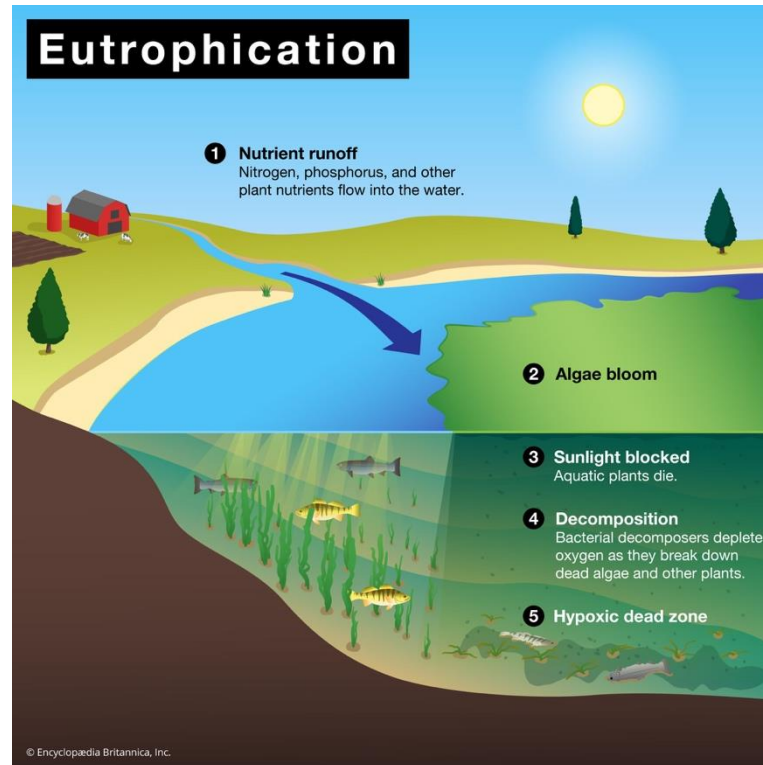
- Databases might underrepresent species
- Sensors need calibration
- Data needs validation



That's where we come in!



“Use It” – Algae-based Fertilizer



Britannica

Eutrophication occurs when extra nutrients end up in waterways.

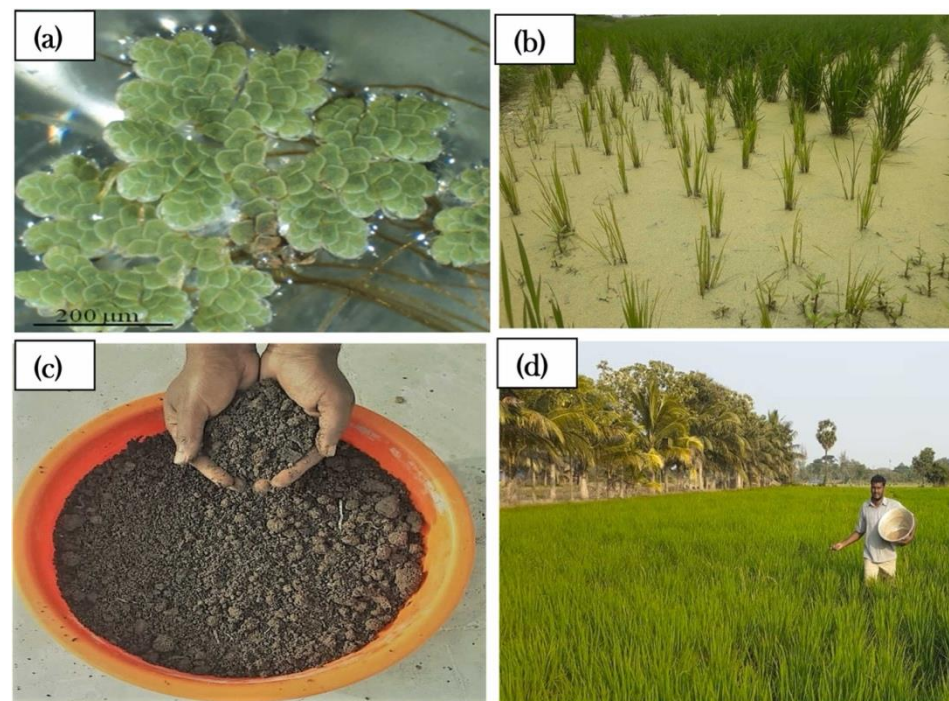
A new ecological niche for microorganisms, especially photosynthetic microbes.

Aquatic habitat destruction...

“Use It” – Algae-based Fertilizer

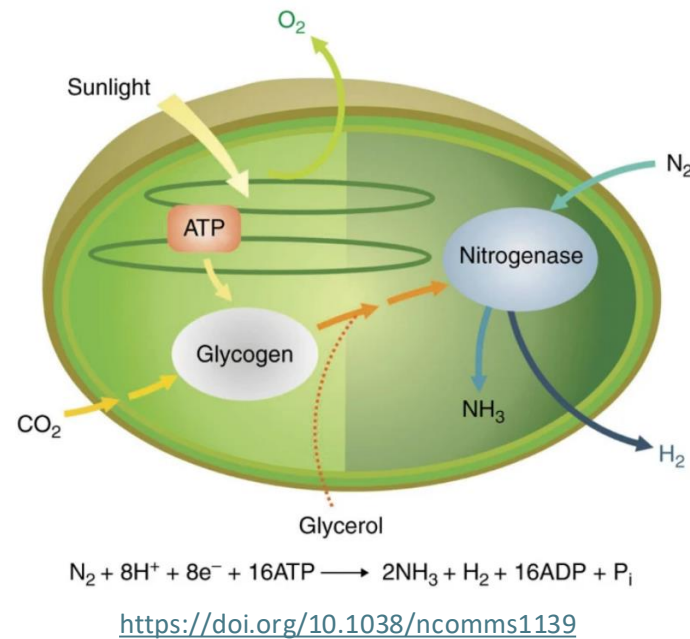
Microalgae are very good at fixing environmental nitrogen into biomolecules.

Azolla macroalgae is commensal with *Anabaena* microalgae



Anabaena azollensis

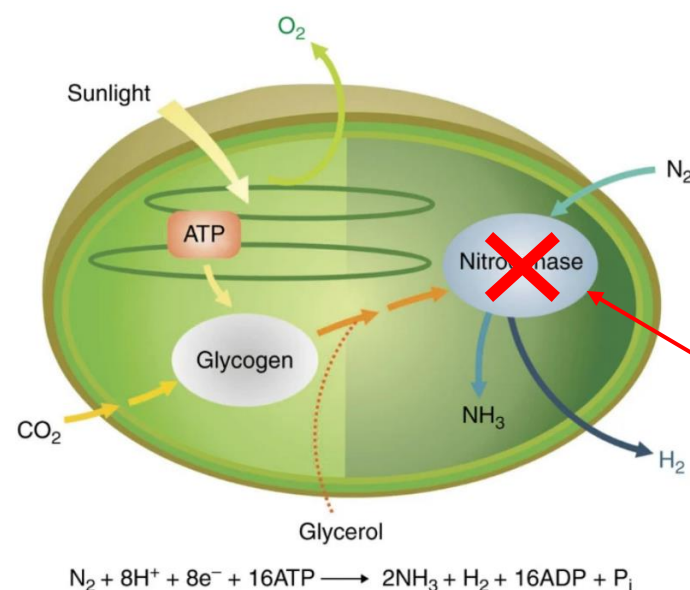
“Use It” – Cyanobacteria Fertilizer



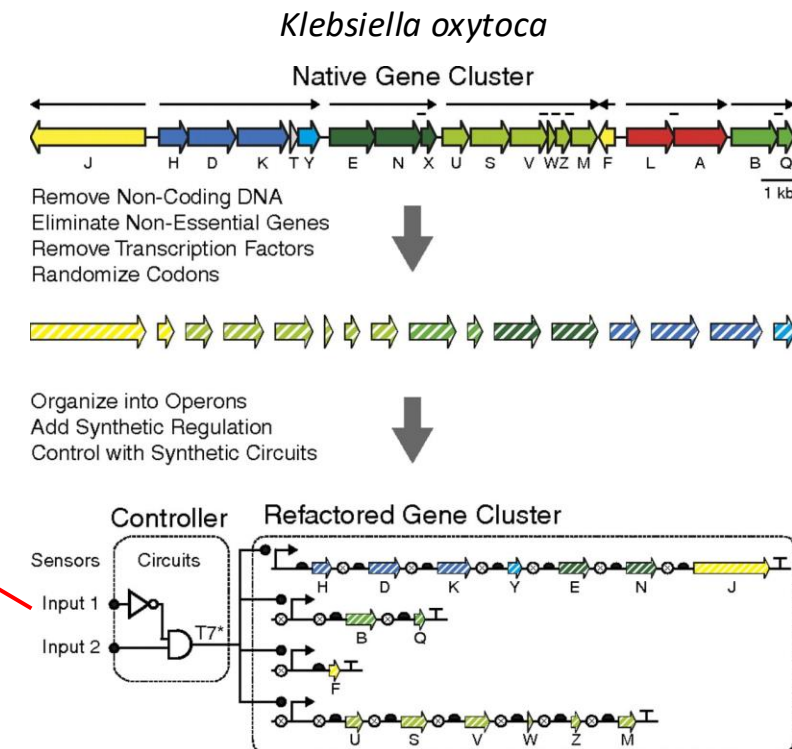
The nitrogenase enzyme is sensitive to oxygen, not good because cyanos make it!



“Use It” – Cyanobacteria Fertilizer



<https://doi.org/10.1038/ncomms1139>



<https://doi.org/10.1073/pnas.1120788109>

“Use It” – Cyanobacteria Fertilizer

<https://www.pivotbio.com/>



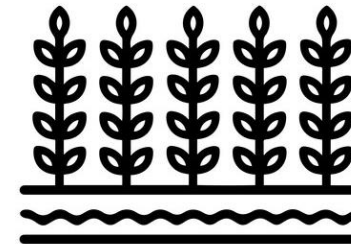
Uses engineered bacteria in the soil to feed nitrogen to growing crops.



Minimized runoff, reduced risk of eutrophication.

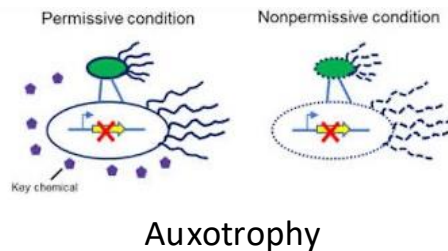
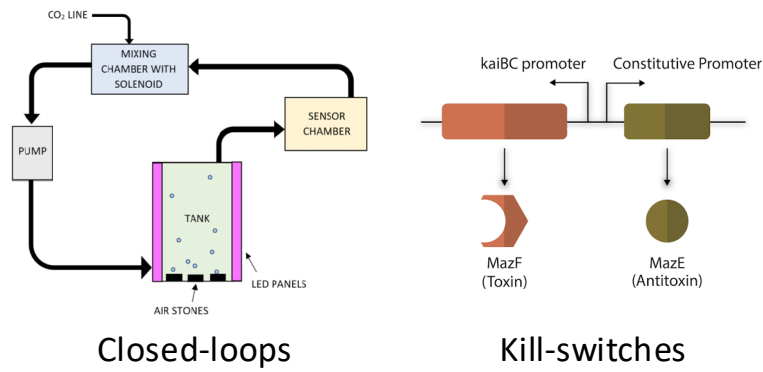
Why not take it a step further with Synthetic Biology?

Imagine crops themselves engineered with the cyanobacterial nitrogen fixation system...



Responsibility in Synthetic Biology

Engineering within living ecosystems



Synthetic Biologists must be responsible for their impact on the environment.

**We should consult people who may be affected.
We should assess possible dual-use of technology.
We should control contamination events.**

Much more...



Responsibility in Synthetic Biology

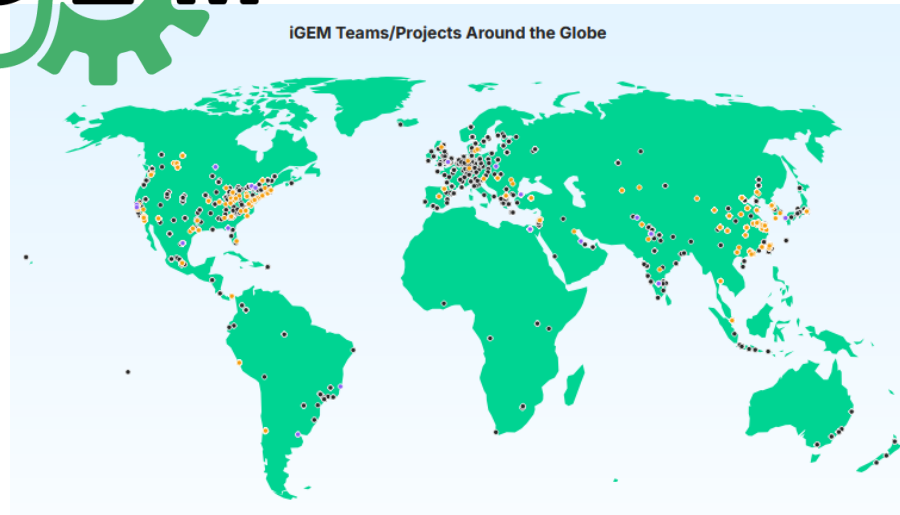
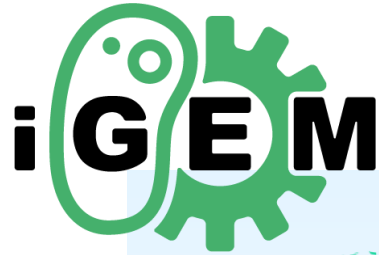
Solutions work best when the communities who live with the problem help shape them.



“Human Practices”



The International Genetically Engineered Machine



>115 thousand iGEMers registered 2004-2026

463 teams compete to solve real problems using *Synthetic Biology*.

iGEM is a global movement to lead synthetic biology into the future.



2022

UBC-Okanagan



2023

ImmuniBee



2026

Evry_Paris-Saclay

iGEM From Above

ME!



Registry Steering Committee



Trevor Fox

Université Paris-Saclay – Évry

France



Participated as **iGEM Competition Judge**
2025 Grand Jamboree
Paris Convention Center, Paris, 28–31 October

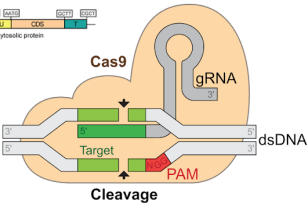
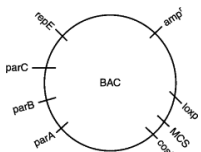
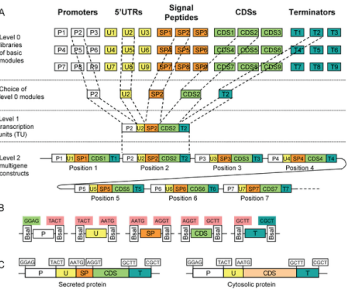
(iGEM Foundation, 2025)

Notes to End On

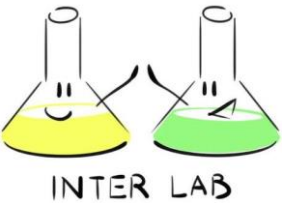
Synthetic Biology is not just about what you do with DNA.

It's also about...

How you do it



Who can do it too



Who you do it for



Thank You



Trevor David Fox

t.d.fox [at] hotmail [dot] com

<https://www.linkedin.com/in/trevdfox/>

